Signet 01

Jim Hannon-Tan, 2026

925 sterling silver, planar antenna, secure element

Foundation Edition · 10 numbered pieces worldwide

The premise

Signet rings have long authenticated documents through a personal mark. Signet 01 carries this tradition into digital communication, where software increasingly prepares and carries out actions on our behalf.

The instrument brings a private signing key and a deliberate gesture together. Touching the ring to a compatible phone signs a specific message or request.

The architecture

The body is investment-cast in 925 sterling silver and finished by hand. The continuous twelve-turn square spiral on its face is both the identifying mark and the working antenna.

Brought close to a compatible phone, the coil receives power and data through NFC. That energy powers the secure element’s signing operation. The ring requires no battery or charging.

Cryptographic intent

The phone prepares the request as EIP-712 structured data and computes its cryptographic digest. The secure element signs that digest using ECDSA and a private key generated and retained on-chip.

A recipient can verify the signature against the signed data and public key. Changes to the signed content are detectable during verification. What the signature authorises depends on the receiving system’s permissions and validation rules.

Technical specifications

Antenna	`12-turn planar spiral · 13.56 MHz`
Secure element	`ARX HaLo · passive NFC`
Signing	`ECDSA / secp256k1 · EIP-712 typed data`
Identity	`LUKSO Universal Profile`

Provenance & The Register

Signet 01 integrates with LUKSO Universal Profiles. The ring's signing key can act as a controller with defined permissions. Those permissions can be changed or revoked on-chain while preserving the profile.

The Register is designed to record each numbered piece's origin, service history and authorised changes of key.

The movement & servicing

The cryptographic movement, the ring's electronic assembly, is designed for studio replacement. The silver body and edition number can be retained, and the surface refinished. A replacement uses a new signing key; the service model records the change in The Register.

Development status. The review interface, Register and service procedures are in development, alongside AI authorisation, blockchain execution, membership and event access.

Concept & industrial design

Jim Hannon-Tan is an industrial designer who has worked in Milan, Amsterdam and London and is now based in Adelaide. His work spans everyday objects, medical technology and digital instruments.

Allocations & contact

jim@jimhannontan.com
www.hannontan.com
ABN 79 609 672 184

New Venture Institute
1284 South Road
Tonsley SA 5042
Australia